

Factorial

The screenshot shows the SIMATS Online Programming Lab interface. The header is blue with the text "SIMATS Online Programming Lab" and "PYTHON PROGRAMMING". A user profile "KUMARAVEL D 192319019RD" is in the top right. The main area is divided into three sections: "PROGRAM EDITOR", "Run" and "Save" buttons, and "INPUT". The "PROGRAM EDITOR" contains a Python script for calculating the factorial of a number. The "INPUT" section shows the number "23".

```
1 num = int(input("Enter a number: "))
2
3 factorial = 1
4
5 if num < 0:
6     print("Factorial does not exist for negative numbers.")
7 elif num == 0:
8     print("The factorial of 0 is 1")
9 else:
10     for i in range(1, num + 1):
11         factorial *= i
12     print(f"The factorial of {num} is {factorial}")
```

output

The screenshot shows the "OUTPUT" section of the SIMATS Online Programming Lab. It displays the result of running the factorial program with input 23. The output text is: "Enter a number: The factorial of 23 is 25852016738884976640000".

```
Enter a number: The factorial of 23 is 25852016738884976640000
```

Max

The screenshot shows the SIMATS Online Programming Lab interface. The header is blue with the text "SIMATS Online Programming Lab" and "PYTHON PROGRAMMING". A user profile "KUMARAVEL D 192319019RD" is in the top right. The main area is divided into three sections: "PROGRAM EDITOR", "Run" and "Save" buttons, and "INPUT". The "PROGRAM EDITOR" contains a Python script for finding the maximum of two numbers. The "INPUT" section shows the numbers "10" and "34".

```
1 a = int(input("Enter first number: "))
2 b = int(input("Enter second number: "))
3
4 if a > b:
5     print("Maximum number is:", a)
6 else:
7     print("Maximum number is:", b)
```

output

The screenshot shows the "OUTPUT" section of the SIMATS Online Programming Lab. It displays the result of running the Max program with inputs 10 and 34. The output text is: "Enter first number: Enter second number: Maximum number is: 34".

```
Enter first number: Enter second number: Maximum number is: 34
```

Palindrome

The screenshot shows the SIMATS Online Programming Lab interface. At the top, there's a blue header with "SIMATS Online Programming Lab" and "PYTHON PROGRAMMING". A user profile "KUMARAVEL D" with ID "192319019RD" is in the top right. The main area is divided into three sections: "PROGRAM EDITOR", "Run" and "Save" buttons, and "INPUT". The "PROGRAM EDITOR" contains a Python script for checking a palindrome. The "INPUT" section shows the number "44".

```
1 num = input("Enter a number: ")
2
3 if num == num[::-1]:
4     print("Palindrome")
5 else:
6     print("Not Palindrome")
```

output

The screenshot shows the "OUTPUT" section of the SIMATS Online Programming Lab. It displays the text "Enter a number: Palindrome", which is the output of the program when the input was "44".

```
Enter a number: Palindrome
```

Area

The screenshot shows the SIMATS Online Programming Lab interface for a program to calculate the area of a rectangle. The header is the same as the previous screenshot. The "PROGRAM EDITOR" contains a Python script that takes length and width as input and prints the area. The "INPUT" section shows the values "10" and "23".

```
1 length = float(input("Enter length: "))
2 width = float(input("Enter width: "))
3
4 area = length * width
5 print("Area =", area)
```

output

The screenshot shows the "OUTPUT" section of the SIMATS Online Programming Lab. It displays the text "Enter length: Enter width: Area = 230.0", which is the output of the program when the inputs were "10" and "23".

```
Enter length: Enter width: Area = 230.0
```

Count vowels

SIMATS Online Programming Lab

PYTHON PROGRAMMING

 KUMARAVEL D
192319019RD

PROGRAM EDITOR

Python

Run

Save

```
1 string = input("Enter a string: ")
2
3 count = 0
4
5 for ch in string.lower():
6     if ch in "aeiou":
7         count += 1
8
9 print("Number of vowels:", count)
```

INPUT

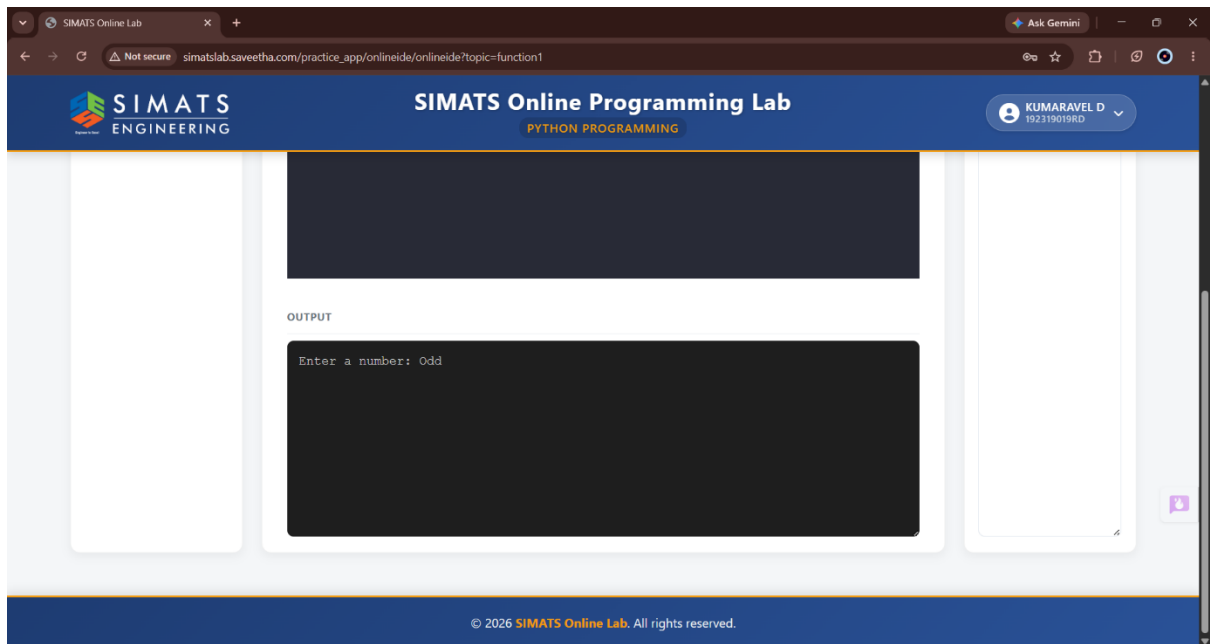
34

output

OUTPUT

Enter a string: Number of vowels: 0

Odd or even



+ -

